

Final Design Report: The Quick-Click Cleaner

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Executive Summary

Many inpatients at the Shirley Ryan Ability Lab in Chicago, IL, utilize ostomy bags while recovering from gastrointestinal surgery or due to diseases such as cancer, IBS, or birth defects. Several of these patients also have fine motor disabilities which make it difficult to perform tasks with dexterity, such as changing an ostomy pouch, which is tedious and requires high levels of precision. To help address this issue, occupational therapist Andrew Morris and nurse consultant Isabelle Brew from Shirley Ryan Ability Lab asked us to design a device that would increase independence for people with fine motor disabilities who use ostomy pouches by simplifying the cleaning/emptying process to be unimanual.

In order to further understand the challenges behind using an ostomy bag both with and without fine motor disabilities, we first met with Andrew and Isabelle to gather more background information about ostomy pouches, ostomy bag routines and procedures, fine motor disabilities, and the emotional effects on users. Afterward, since an in-person user observation was not possible, we watched two videos by the University of Chicago School of Medicine on how to change and care for an ostomy bag. After building three mockups, we met with Andrew and Isabelle again to receive their feedback and suggestions on our initial ideas. Then, we designed a three-part system and independently conducted an initial testing to see if our prototype worked and whether we needed to make any final adjustments.

Our design, the Quick-Click Cleaner, is a three-part system. The first—and main—device in the system is the cleaning wand, which consists of a handle and molded head that perfectly fits around the inside and outside of the ostomy pouch opening. A disposable cotton insert is locked into the head before cleaning, and can be easily ejected with a spring-loaded button once soiled. This design makes it easy and quick to clean the inside and outside of the bag, which has small corners and can be difficult to reach, as well as reducing the mess typically associated with changing one's ostomy pouch.

The second and third parts of the Quick-Click Cleaner are the disposable cotton insert and base. The cotton insert, like the head, is molded to fit around both the inside and outside of the ostomy pouch opening so that a user can clean both sides simultaneously. The base stores fresh cotton inserts and also acts as a device to help the user attach the cotton inserts.

Although our device was designed specifically for bedridden people with fine motor disabilities, with further user testing and developments such as reducing the size, the Quick-Click Cleaner has the potential to provide many people with fine motor disabilities an easier and cleaner way to clean their ostomy pouches and thereby increase their independence.

Introduction

The process of cleaning an ostomy bag requires high levels of dexterity and caution. For this reason, people with fine motor disabilities often find it difficult to clean their ostomy bag. Specifically, cleaning an ostomy bag requires the user to keep the bag open with one hand while simultaneously using their other hand to squeeze out the waste. Further, this position must be maintained for several minutes in order to clean the bag. This process not only requires fine motor skills, but it also requires the user to sit on a toilet and clean their bag with their bare hands. Many users find cleaning the bag to be uncomfortable as it requires them to interact directly with their waste, and for users with fine motor disabilities that are bed-bound, the process of having to walk to the toilet is physically challenging.

Throughout the decades, the design of ostomy bags has evolved to tackle issues of leakage, odor, visibility, and application. However, solving these problems has not involved the population of ostomy users with fine motor disabilities, making it virtually impossible for them to independently empty their bags. More generally, the issue of having to interact with one's own waste has not been addressed either even though it affects all ostomy bag users. Due to this, creating a product that allows patients in Shirley Ryan Ability Lab with fine motor impairment and limited mobility to clean their ostomy bags effectively and comfortably is crucial.

The Quick-Click-Cleaner tackles all the issues listed above. The device resembles the shape of a lint roller with a cylindrical top. Specifically, the top holds a disposable cotton insert which serves as the cleaning head. The unique shape of the cleaning head (a cylinder within a hollow cylinder) will allow the cotton insert to simultaneously clean the inside and outside of the bag. The base of the device is made out of PLA that has a built in spring ejection mechanism that disposes of the soiled cotton insert with just the push of a button. This button is located at the end of the base. During use the user will hold onto the handle of the cleaning wand which means their hand will be further away from the bag, thus avoiding any messy or uncomfortable interactions with waste. In addition, when the user pushes the button the soiled cotton insert will be automatically disposed of and a new one will be pushed to the top. Overall, the design minimizes the level of dexterity needed to clean an ostomy bag and does not require the user to be in a bathroom since they can keep Quick-Click-Cleaner near their nightstand.

This report discusses how Quick-Click Cleaner directly tackles the issues of ostomy bags being created with only physically-abled people in mind. We offer our detailed procedure for assessing the target's users needs and how our device resolves all of the key challenges faced by our client. Furthermore, we discuss the restrictions of our current design and how we will address them in the future to improve our current design.

Users and Requirements

Users

Primary Users

Our product is designed for ostomy bag users that have fine motor disabilities which make cleaning an ostomy bag challenging for them. The goal of our product is to simplify that process by creating an external tool that can simultaneously clean the outside and inside of the ostomy bag unimanually while minimizing mess. Given that our client has fine motor issues, the cleaning device's ejection feature will be able to make it easier for the user to remove the dirty tissues and the duck-bill tongue makes inserting the device easier.

Secondary Users

While our product is specifically designed for people with fine motor disabilities that are bedridden, the design is also portable. This means that it can be taken on the go in an already existing ostomy care bag and be used by the general ostomy bag wearing population to aid in the sometimes uncomfortable act of cleaning their ostomy bag.

Caretakers

Currently, most ostomy bag users with fine motor disabilities have another person help them clean and empty their bag. This product can then help give the target users a greater sense of independence and a greater ability to be self-sufficient.

All Ostomy Bag Users

Beyond people with fine motor disabilities that are bedridden, the tool is designed to make the difficult and messy task of cleaning the opening of an ostomy bag easier and more sanitary. The mechanism and ease of use will still be helpful for all ostomy bag users.

Requirements

Minimizes Use of Fine Motor Skills

The primary users have fine motor disabilities; as such, the device cannot require a high level of dexterity or other fine motor skills and must be easy to manipulate.

Cleanliness

The Quick-Click Cleaner has to be clean and separate the person from the fecal matter they are cleaning. Specifically, it needs to keep the person's hand at a minimum, 3 inches away from the opening of the bag and not require the person to touch the soiled tissue at any point in the process.

Storage

The Quick-Click Cleaner must have a storage base that is easy to access. As our primary target user is bedridden thus they need to have a place to put the device when they are not using it.

Design Concept and Rationale

The Quick-Click Cleaner is a handheld cleaning wand compatible with any drainable ostomy pouch with a velcro/clamp closure (see Figure 1). The device consists of three parts; first, the base, which holds the cleaning wand while not in use and stores the cotton inserts (see Figure 2). Second, the disposable cotton inserts are attached to the cleaning wand to clean the inside and outside of the ostomy bag simultaneously (see Figure 3). Finally, the cleaning wand consists of a handle and ostomy pouch opening-shaped head to which the cotton insert can be easily inserted and ejected. In order to eject the cotton insert without the user needing to touch the soiled cotton insert after usage, the cleaning wand has a spring-loaded mechanism within the handle to easily allow for the ejection of the cotton insert with the push of a button. Moreover, loading a fresh insert is also easy, as it only requires a new one to be pressed into the cleaning wand's head. These three parts of the Quick-Click Cleaner work together as one cohesive system to optimize the cleaning process of the ostomy bag for bedridden users with fine motor disabilities. Specifically, the Quick-Click Cleaner can be used unimanually and allows users to independently clean their ostomy pouches while also minimizing the mess associated with cleaning an ostomy pouch.



Figure 1: Quick-Click Cleaner in use



Figure 2: Base



Figure 3: Disposable paper insert

The following sections describe the components of each part of the system and the rationale for each component.

Part 1: The Cleaning Wand

Handle

The handle of the cleaning wand is a cylindrical round grip made of PLA that runs through the middle of the wand (see Figure 4). It is 3 inches long and has an outside diameter of 3 inches. The handle is easy to grasp and smooth all around. On the top of the handle is the cleaning head and on the bottom of the handle is an ejection button to release a soiled cotton insert. The handle contains the ejection mechanism, which is further discussed below.



Figure 4: Handle of the cleaning wand

Rationale

During the client interview (see Appendix B), the clients emphasized that users of our product would have fine and/or gross motor disabilities, preventing fine, precise movements with the fingers. They added that people with fine/gross motor disabilities often find it easier to grasp larger objects, rather than smaller objects. Also, during the user observation (see Appendix D), we noticed that the cleaning of the interior and exterior of the ostomy pouch opening required a lot of dexterity and small movements of the fingers. The cleaning of the ostomy pouch was also quite messy. We therefore designed the handle to be large and easier for the user to handle, in addition to being relatively long such that the mess associated with cleaning an ostomy pouch would be mitigated.

Cleaning Head

The main component of the cleaning wand is the cleaning head, which is a cylinder with a hollow opening and a smaller cylinder inside (see Figure 5), to allow the user to easily clean the interior and exterior of their ostomy pouch opening simultaneously. The diameter of the outer part of the head is 2 inches long, the diameter of the inner part of the head is 1.5 inches long and it is 2 inches tall. The inner cylinder, which has a duck-bill shaped tongue to allow the user to easily insert it into their ostomy pouch, is 1 inch taller than the outside cylinder. Disposable cotton inserts can be easily inserted into the cleaning wand's head and a spring-loaded mechanism allows them to be quickly ejected after use.



Figure 5: Cleaning head

Rationale

During the client interview (see Appendix B), the clients mentioned that one part of the cleaning process that was difficult was cleaning the corners of the pouch, which are narrow and therefore difficult for people with fine motor disabilities to access. As such, the shape of the head was designed to reach all sides and corners of the ostomy pouch opening simultaneously while only requiring vertical and twisting motions with the wrist, rather than the fingers. Moreover, the cotton inserts easily access all sides and reduce mess, since they are shaped to the cleaning head and disposable; messiness was another aspect of the cleaning process that many ostomy bag users disliked, as mentioned in the client interview (see Appendix B).

Spring-loaded ejection mechanism

The spring-loaded ejection mechanism is made up of a spring contained within the handle that is connected to a button at the bottom of the handle and an ejector head (see Figure 6). The spring

is 1 inch long with a spring constant of 4 Newton/Meter. In the down position, the cotton insert is locked into the cleaning head (see Figure 7). In the release position, the top-component of the handle springs up and releases the soiled cotton insert for easy disposal (see Figure 8).



Figure 6: Cross-section of cleaning wand



Figure 7: Ejection mechanism in the cleaning position



Figure 8: Ejection mechanism in the eject position

Rationale

During both the client interview (see Appendix B) and the user observation (see Appendix D), we noticed that cleaning the ostomy pouch opening was messy and required numerous tissues and/or squares of toilet paper. To mitigate the mess, we created a mechanism that allowed the user to not have to touch the soiled cotton insert that could be operated unimanually.

Part 2: Disposable Cotton Insert

The disposable cotton insert is shaped around the interior and exterior of the ostomy pouch opening (akin to the cleaning head) and is made out of cotton (see Figure 3). It is inserted into the cleaning head, used until soiled, and then disposed of in the garbage. Each Quick-Click Cleaner will come with 25 disposable cotton inserts (enough for approximately a week's worth of cleanings), with refills available.

Rationale

In order to optimize the cleaning process, we opted to create a disposable cotton product that was specifically shaped to the ostomy pouch opening to simultaneously clean the inside and outside of the opening. During both the client interview (see Appendix B) and the user observation (see Appendix D), we noticed that cleaning the ostomy pouch opening was messy and required numerous tissues and/or squares of toilet paper. While toilet paper or wipes may have been a viable option, they would have been less effective in conjunction with our device; as such, while it is of further cost to the user, we believe that the ease of use and effectiveness of the disposable cotton insert will outweigh the potential extra cost, as evidenced by the performance testing we conducted (see Appendix H).

Part 3: The Base

The base of the Quick-Click Cleaner is made from PLA and it is 0.5 inches tall x 5 inches wide x 5 inches long (see Figure 2). The base has a hollow cylinder 3 inches tall with an inner radius of 0.75 inches and outer radius of 0.85 inches to hold the Quick-Click Cleaner when not in use. The cylinder on the base also allows users to load the cotton inserts into the cleaning head.

Rationale

During the client interview (see Appendix B), we learned that our clients worked primarily with bedridden people and people in wheelchairs. Since we decided early on that our device would not be portable and for on-the-go use, we opted to focus on designing for bedridden people; we wanted them to have a way to both store the device and reload the device on a bedside table. The base needed to be sturdy enough to hold the Quick-Click Cleaner when not in use as well as withstand pressure when a new cotton insert was loaded.

Directions for Future Development

Our design as it is helps to simplify the process of cleaning an ostomy bag, but it can be improved in the following ways:

Testing

Performance Testing

The Quick-Click Cleaner was designed to withstand common physical stresses including regular usage multiple times a day, moderate to heavy pressure applied on the handle and hand, and being dropped on the ground from bedside height. However, the exact metrics of the device's physical limitations are unknown. We did some initial performance testing, but further performance testing of the Quick-Click Cleaner would help establish boundaries of use and extend its use to a broader range of users.

User Testing

The Quick-Click Cleaner was designed without a specific primary user in mind, but rather a specific group of people: bedridden people with fine motor disabilities. However, as the device continues to develop, extending its use to a larger range of users could provide more people with fine motor disabilities (bedridden, confined to a wheelchair, or ambulatory) and people who are bedridden (without fine motor disabilities) the possibility of independently cleaning their ostomy pouch. Such a device could have significant emotional and physical benefits for both the user and their caretaker. In order to accomplish this, we recommend conducting further user testing of the device with people not within our initial group of primary users to see how its effectiveness may change.

Improvements to the Design

Shape of Head

The shape of the head of the Quick-Click Cleaner was specifically designed to allow the opening of the ostomy bag to fit within the crevice. However, due to time constraints, the shape of the head was not perfectly optimized to fit the ostomy bag; currently, not all sides of the ostomy bag can be cleaned simultaneously with one in/out motion. For this reason, a potential improvement to our design would be to adjust the shape of the head such that the user would only have to insert and remove the Quick-Click Cleaner without having to twist the device around to clean all sides of the ostomy pouch opening.

Size of Quick-Click Cleaner and Removal of Base

During the design phase of our project, we decided to focus on bedridden users. As such, the amount of space the Quick-Click Cleaner and its base took up was relatively insignificant (as long as it fit on a bedside table). However, if the size of the device were smaller, it might be able

to reach a broader audience, such as people in wheelchairs with fine motor disabilities, who could store a smaller device on their wheelchair. Ideally, the base would be an optional addition to further reduce space (this, however, would also present an additional issue of having to figure out a new way to uni-manually attach the insert). Due to time constraints, we did not attempt to scale the device down; however, we believe that a potential improvement to the design would include additional device sizing and an optional base for users in wheelchairs or even for day-to-day use by people without fine motor disabilities.

Conclusion

To summarize, our device's design meets the essential requirements and needs of patients with fine motor disabilities who utilize ostomy bags. It successfully assists patients by making the cleaning process smoother and less uncomfortable. Our device utilizes:

- a 4 inch long solid handle that is comfortable for users of all hand sizes to easily hold and move around
- an easily pressable, large spring-loaded button for quick ejection of tissue with a spring constant of 4 N/m
- a cylindrical cleaning section with an inner radius of 0.5 inches and outer radius of 1 inch that is wide enough for the ostomy bag opening to easily slide in and allows for cleaning both inside and outside of the bag.
- an ejectable disposable cotton insert that is used for comfortable cleaning and swabbing of the opening

Ostomy bag users with fine motor disabilities need an ostomy bag innovation that simplifies the cleaning process as it is difficult, requiring significant preparation and precise movements. The cotton inserts clean the waste with ease, while the double layer of cylinders allow for cleaning of both the inside and outside of the bag simultaneously. Furthermore, patients need to have a reduced number of steps for cleaning to minimize the possibility of accidents. This device meets this with a straightforward cotton insert and a simple button ejection system.

The device will be a one-time purchase for patients where people can be able to reuse the device for each clean. Factoring in the print and spring costs, the device will cost approximately \$25 to manufacture. The cotton inserts will have a similar cost to purchasing toilet paper, so it will not require extra costs. The low manufacturing costs for the cotton inserts means the cotton inserts can be easily made and purchased by patients in the long term.

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Appendices

Appendix A: Background Research Summary

At the beginning of the project, we did background research on terms and concepts associated with ostomy pouches and stomas. Our project involves designing a method to allow people with fine and gross motor disabilities to uni-manually empty ostomy pouches. While ostomy pouches are not common, there is still a significant population who both use ostomy pouches and have fine/gross motor disabilities, often due to stroke, tremors, arthritis, spinal cord injuries, and other diseases. Our background research allowed us to begin to understand the several aspects of the problem such that we were prepared to ask informed questions during our client interview. In particular, we researched: (1) basic information about ostomy pouches and the reasons why someone might be using one; (2) basic information about fine motor disabilities; (3) the steps involved in emptying and changing an ostomy pouch; (4) current solutions for people with fine motor disabilities; and (5) the fine motor skills needed to replace an ostomy pouch.

Ostomy pouches and stomas

An ostomy is an artificial opening in an organ of the body, created during an operation, for either temporary or permanent use. An ostomy allows bodily fluids and waste to be redirected through a stoma—an opening—into an ostomy pouch—a heavy duty plastic bag worn on the body. When someone's urinary or digestive system is malfunctioning, their bodily waste must be rerouted from its usual path to avoid complications. Thus, an ostomy bag becomes the only way for the person to be able to heal the part of the body that is malfunctioning, or in more severe cases, to replace the function of that organ system (Cleveland Clinic, 2021).



Figure 9: Ostomy pouch

Some common diseases that lead to having an ostomy bag are bowel cancer, IBSs, birth defects and even severe abdominal or pelvic trauma from injuries (such as gunshot wounds or knife wounds). Procedures that would require an ostomy bag include a colostomy (the surgical

removal of part of the colon or rectum), urostomy (the diversion of urine away from a diseased or defective bladder), or ileostomy (the surgical removal of the lower part of the small intestine) (United Ostomy Associations of America, n.d.).

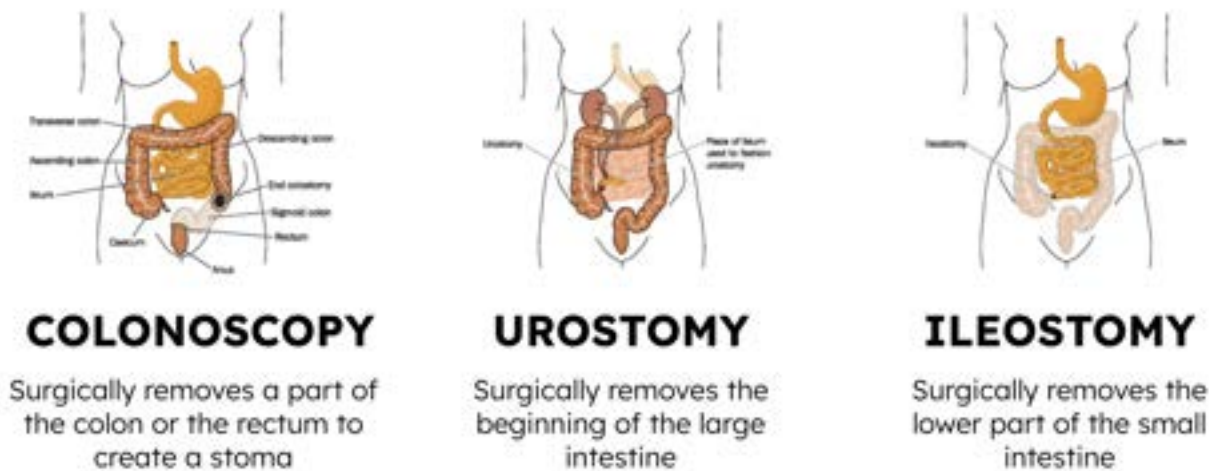


Figure 10: Colostomy, urostomy, and ileostomy stoma sites

There are several ostomy bags on the market, all of which fall into one of two categories: a one piece pouch system with just the bag and skin barrier, or a two piece pouch system with two layers of skin separation from the bag. All in all, there are approximately 30 different types of bags for different body types to customize for weight, height, posture, activity level, and location of stoma (Washington University, 2022).

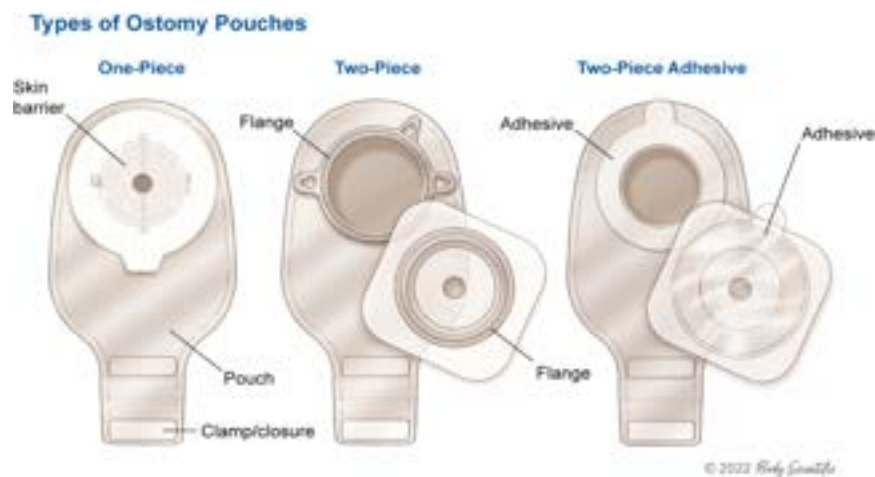


Figure 11: Parts of an ostomy pouch

Fine motor disabilities

A fine motor disability is the inability for someone to perform tasks that require some level of manual dexterity, which is the using of hands to manipulate objects and perform precise movements (Burr, 2022). Fine motor disabilities are often symptoms of an underlying disease

rather than a disease in and of itself. Elderly people often have fine motor disabilities as a result of aging and the weakening of muscle use, though young children who are still developing can also have fine motor disabilities as a result of lack of practice with actions involving manual dexterity (Visiting Angels, n.d.).

Steps to empty and change an ostomy pouch

There are several steps to empty and change an ostomy pouch. (For the sake of this summary, we will be focusing on the two piece pouch system.) Beginning with emptying an ostomy pouch, which typically occurs multiple times a day depending on the location of the stoma and buildup of waste in the pouch, the opening at the bottom of the ostomy pouch is unrolled and the waste is put into either a bag or a toilet. Then, the opening is wiped clean, rolled back up, and typically secured with velcro or some other straps (United Ostomy Associations of America, 2016)

To change an ostomy pouch, which typically occurs every 1-3 days, a wipe is first used to remove the adhesive holding the ostomy pouch, before gently removing the pouch from the skin. The stoma and surrounding skin is then cleaned and then an ostomy barrier seal is placed on the stoma. After, the pouch flange is placed carefully around the stoma and the ostomy bag is then locked into the flange (Phoenix Children's, 2017). (For further information and graphics on the emptying and cleaning of an ostomy pouch, see Appendix D.)

Current solutions for people with fine motor disabilities

There are currently some solutions and therapies to help people with fine motor disabilities maximize their independence in day-to-day activities. Most of them involve practice for children to help develop and maintain their fine motor skills, such as playing with bubbles or small toys, but the elderly can also benefit from such therapies, such as finger painting or putting on jewelry. There also exist products that minimize the need for fine motor skills by decreasing the degrees of freedom and increasing the room for error (for example, utensils with bigger grips) (NAPA Center, 2023).



Figure 12: Fine motor disability therapies

Fine motor skills and emptying/changing an ostomy pouch

There are several steps to emptying/changing an ostomy pouch that involve the use of fine motor skills. First, emptying, folding, closing, and locking the ostomy bag closure requires strength and gross motor skills. Also, there needs to be control of the fingers and hand to gently remove the skin barrier without damaging skin or the stoma. Holding a wipe or adhesive with fingers when one has a fine motor disability could be difficult as well. Often, the ability to use scissors is particularly important as the barrier ring needs to be cut to perfectly fit the stoma, which is typically irregular. Finally, even the simple act of screwing/unscrewing the cap of tubes of adhesive can be difficult, as they are often small and hard to grasp (Baylor College, n.d.).



Figure 13: Velcro flap on Hollister ostomy pouch



Figure 14: Barrier ring



Figure 15: Tubes of adhesive

Appendix B: Client Interview Summary

Introduction

Through this research, we are trying to confirm the general background knowledge regarding ostomy bags and trying to learn about some specifics or highlighted issues that the client would like the project design to focus on.

Methodology

We are performing this research through an online interview through Zoom. This will be conducted with other project groups so that more varied or specific questions can be answered.

Results

Specific Targets

- Focus on innovating not inventing ostomy bags
 - Ex: Hollister Ostomy Bag
 - Odor Prevention
- Target solid waste (fecal) over liquid waste
 - Solid waste is significantly more difficult to expel and requires to steps
 - Solving solid waste problems will subsequently solve liquid waste problems
- One & Two Pouch Systems
 - Do not emphasize trying to improve specifically one or two pouch systems, but rather systems that affect ostomy bags as a whole

Motor Skills

- Fine Motor: Fine, precise movements with your fingers
- Gross Motor: Moving hand in space and holding hand up in a general location
- Manual Motor: Moving one side of the body

Common Steps that Require Fine Motor Skills

- Peeling off adhesive
- Pasting adhesive paste onto skin barrier
- Stretching O-rings to fit stoma
- Cutting holes on pouches to appropriate size with scissors
- Folding and pressing velcro
- Pushing sides of the bag to empty out the bag
- Screwing/Unscrewing caps to expel

Daily Routine

- Emptying out bag on an average of 8x daily
- Change ostomy bag completely 2-3x a week
- Emptying out is done over a toilet or basin
 - Varies in height per person depending on mobility
- Carry extra ostomy bags when going outside
 - In case of emergencies

Personal Complaints or Considerations

- Skin Irritation during peeling
- Leakages of liquid or solid waste
- Odor leakages that result in smell of person
- Bag Bursts
- Stoma Placement
 - Position of bag during situations like wearing seatbelts, etc.

Summary

Through this design project, we should focus on innovating the ostomy bag rather than attempt inventing a completely new system of ostomy bags. This means that we should look at specific details and functions of the ostomy bag, identify possible motor issues that could occur during the process of using that function, and look for possible solutions to fix and resolve that issue.

An example of an innovation of the ostomy bag includes the Hollister ostomy bag which focuses on odor and other issues regarding expelling. Following this example, we should attempt to innovate small issues. Another focus we should shift to was regarding the type of ostomy bag we should work to develop. There are two different types of waste in ostomy bags, solid and liquid. Because solid waste is much more complicated and troublesome to expel from an ostomy bag due to having to possibly shake bags to get the waste out and the expulsion movement being less predictable from solid waste, we should focus on the harder issue at hand. Furthermore, by fixing issues regarding solid waste, it can subsequently help solve issues with liquid waste.

On top of the waste inside the ostomy bag, there are different types of ostomy bags that have different methods of expelling and changing. Specifically, there are one and two pouch systems. We should not try to focus on what type of bag but rather the motor skills that are required for general ostomy bags.

To expel waste from an ostomy bag, different types of motor skills are required. There were three specific types of motor skills discussed in the interview: fine, gross, and manual. Fine motor skills include fine, precise movements with your fingers. Gross motor skills involve holding your hand up in space and placing it in a general location. And manual skills are those that involve movement of one side of the body.

Some common steps that involve fine motor skills in changing and emptying an ostomy bag include peeling off adhesives, stretching and fitting O-rings, cutting holes in pouches to appropriate size, pulling tabs to expel, folding and pressing velcro, pushing sides to empty out the bag, and screwing/unscrewing caps. In order to resolve these problems, we should aim to change the bag by allowing these steps to be performed with only one side/hand of the body.

It is also important to consider a person's daily routine when designing the product and how to use it. For example, emptying an ostomy bag occurs an average of 8x per day, and changing an ostomy bag happens 2-3x a week. When ostomy bag users empty a bag, it is done over a toilet or basin that can be located at several different heights based on a person's mobility. During the changing and expulsion process, issues such as leakages, irritation to skin, odor smells, and bursts can occur as well. Other considerations such as noting that people carry extra ostomy bags on them when going outside or different stoma placements is important as well throughout the design process.

Discussion

From this interview, we learned that the design of this ostomy bag for people with fixed motor disability should aim to incorporate innovations that allow one-handed steps during the expulsion and changing process. It was also noteworthy that we can also brainstorm in other dimensions or perspectives such as developing tools to help change bags or attaching another device onto the bag. Ultimately, we want to minimize the degrees of freedom needed to operate the ostomy bag to increase stabilization.

Appendix C: Project Definition

Project name: Unimanual Ostomy Bag Cleaner

Client: Andrew Morris and Isabelle Brew

Team Members: Lauren Ho, Melanie Cuenca, Barkot Simegn, Edward Lee

Date: November 16th, 2023

Mission Statement

Create an external device for bed-bound ostomy bag users with fine motor disabilities to comfortably and efficiently clean their ostomy bag.

Team Deliverables

The final report, presentation and a working prototype

Constraints

- Less than 10 weeks to understand, brainstorm and build a working model that will solve the clients issue
- A budget of \$150 for materials and other sources that would be used to create a working model
- The external tool has to be changed and put back together for consecutive use using only one hand
- Can clean the inside and outside of the bag simultaneously
- Target solid waste over liquid waste
 - Solid waste requires more fine-motor skills
 - Solving solid waste problems will subsequently solve liquid waste problems
- Use the Hollister bag as the model

The target user is a patient in Shirley Ryan Hospital who struggles with tasks that require a degree of dexterity. In addition, they spend most of their time in bed with not much mobility, thus they require assistance to change and clean their ostomy bag. The assistance nurse comes in as usual to help the patient clean their ostomy bag. They prepare toilet paper, a bin, and gloves. They prop the patient up and begin emptying the waste into the bin. The patient has done this countless times before, yet it is still uncomfortable to have someone else clean your waste for you. They have to do this around 3-6 times a day.

Users/Stakeholders

- People with fine-motor impairments and use an ostomy bag (target users)
- People with ostomy bags in general
- Shirley Ryan Hospital (the institution that placed a request for design improvements of ostomy bags for their patients)

Users Profile

The target user is a patient with fine motor disabilities meaning they struggle with tasks that require a degree of dexterity, they also are bed-bound and use a Hollister ostomy bag.

Illustrative User Scenario

The user is created using the information we were given by our client Andrew Morris and his experience assisting ostomy bag patients in Shirley Ryan Hospital as well as research of the daily life of ostomy bag users with fine motor disabilities

The user wakes up and they stay on their bed until their assistance comes to check their condition and their ostomy bag. The assistance shows up prepared with a bin, gloves and rolls of toilet paper. As much as they would like to clean themselves up the user struggles with keeping the bag open as that requires using both of their hands and rolling toilet paper around their hands which is challenging since they have weak motor skills. The nurse props them to the edge of the bed and begins emptying the waste inside into the bag. Once it's done the nurse grabs enough toilet paper and begins cleaning inside and outside the bag to make sure the bag is thoroughly clean. Even though the user has done this several times, the intimate nature of this procedure makes them feel uncomfortable and icky since their assistant interacts directly with their waste. They repeat this process 3 to 6 times a day.

Table 1: Project Requirements

Needs	Metrics	Units	Ideal Value	Allowable Value
Can easily be gripped/used one hand	Max Length	Inches	2	2
	Max Height	Inches	6.5	6.5
	Max Width	Inches	2	2
Can stay put on a night stand near the users bed				
The users hand does not directly interact with the bag				

Needs	Metrics	Units	Ideal Value	Allowable Value
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Efficiently eject the used cleaning material and recharged with a spring mechanism	Overall Length	Inches	1	1
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Needs	Material	Units	Ideal Value	Allowable Value
Cleaning material	Compressed cotton balls	A pack is going to have 6 cleaning materials	6	1

Appendix D: User Observation Summary

Methodology

We watched 2 videos, “Ostomy Bag Pouch Change | Ostomy Care Nursing | Colostomy, Ileostomy Bag Change” by Nurse Sarah on Youtube, and “How to Empty an Ostomy Pouch” by the University of Chicago’s School of Medicine. Between these videos we have grabbed 9 images that demonstrate the process in total for emptying and changing a bag, and all the important choke points useful for analysis.

Pictures and Analysis



Figure 16: Man unclipping an ostomy bag

In Figure 16, we can see a man unclipping an ostomy bag opening to empty it. The specific clip he is using isn't the only way to open or close the bag, it is one of many including velcro bags and high output bags. All of these bags present the issue of difficulty opening with a fine motor disability or unimanually.



Figure 17: Man positioning ostomy bag

Figure 17 shows how hard it can be to aim an ostomy bag. Without even emptying the bag fully, he needs 2 hands to be able to aim the bag. Furthermore his hands are very close to the excrement, which is very unhygienic.



Figure 18: Man cleaning ostomy bag with tissues

Figure 18 demonstrates the traditional cleaning process and how wasteful and unhygienic it is, as the amount of toilet paper it takes to be able to clean this bag is substantial. This goes to show how bad the current process is.



Figure 19: Man cleaning ostomy interior seal

Figure 19 is showing the man cleaning the interior, again with a thick wad of toilet paper. In order to do what this man is doing, you have to have good dexterity. If you have a fine motor disability or have one hand, you couldn't do this.



Figure 20: Tools used to clean and change an ostomy bag

Figure 20 shows how much work it takes to just open the wafer of an ostomy bag. It is very important to simplify this process to be able to allow people to do this on their own, even with limited skill or physical ability. Some obvious points of difficulty are using a stencil and pen to get a circle onto the bag, cutting the wafer, peeling the adhesive, etc.



Figure 21: Simulated redness from an improperly sealed ostomy bag

Figure 21 shows what can happen if the seal of an ostomy bag isn't perfect. The redness in the image comes from an infection that happens from the moisture and bacteria trapped within the seal due to a poor seal.



Figure 22: Cleaning in and around the stoma in preparation for a replacement bag.

Figure 22 shows the process it takes to clean the stoma and prepare it for a new bag. This helps give us an idea of the other side of the ostomy bag and gives us an idea of how what is already being done to clean in and around the ostomy bag and stoma is already done to give us a better idea of what can be improved upon rather than starting from scratch.

Table 2: Quotations and Analysis

What We Saw:	What Was Said About it
● Figure 16: ○ Unclipping the bag to open it ● Figure 17: ○ Squeezing the bag out when the contents were stuck ● Figure 18: ○ Cleaning the bag with toilet paper ● Figure 19: ○ Using toilet paper to clean the interior and exterior until there is no residue left	● There are multiple clip systems that exist to allow flexibility in this aspect ● This was described as a means to clean and empty the bag and get out the hard to get things ● This was because the outside needs to be wiped and cleaned every time the bag is emptied according to the doctor. ● This is because not only does the very tip need to be clean, any part of the bag that will be exposed to the outside needs to be clean.

● Figure 20: ○ Everything you need to change an ostomy bag ● Figure 21: ○ Redness around the stoma ● Figure 22: ○ Cleans both the outside and inside of the stoma	● These include, clean towels, gloves, scissors, templates, and a marker, all used to make the experience as precise, not as easy as possible ● She says it's probably due to an infection, the bag needs to be sealed better. ● To remove the adhesive, any poop/sweat/etc buildup on the skin, and clean the area for a new pouch
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Links to the videos used for user observation:

<https://www.youtube.com/watch?v=h8CtsPAaa5Y>

<https://www.youtube.com/watch?v=NVrO-jITUTQ>

Appendix E: User Feedback and Testing Summary

Introduction

Our meeting was conducted on Zoom on October 30th, 2023. This meeting was with our DTC group and the clients, who are working with patients with bags. In this meeting, we presented the various brainstormed ideas we had surrounding how to improve or innovate the ostomy bag for patients who have fine motor disabilities. We utilized mockups, visuals, gestures, and motions to present the ideas effectively to the clients.

Bag Lever Opener

This idea was using a relatively large lever horizontally attached to the ostomy bag opening that patients could push with their palm in order to open the closed opening of the bag. It would utilize a mechanical advantage of a lever arm in order to reduce the amount of strength or fine hand movement in order to open an ostomy bag. It would be removable from the bag in order to reduce costs and apply reusability. The client did not see much viability in the lever design as the lever was too large in proportion to the ostomy bag, which could obstruct the patient's day to day life. Furthermore, the reusability and attachability of the lever arm onto the ostomy bag was not visualized in its entirety, making it more difficult to imagine using this device.

Cleaning Tool

The cleaner tool is a separate device that would be used to clean the inside of the ostomy bag's opening. This cleaner tool functions similar to a lint roller. The device would have two sections - a top and a bottom section. The bottom section is a long rod which will be used for holding the device with an entire hand. This rod being able to be grasped will reduce the need for fine motor skills and allow patients to easily hold the device. The top section will be a slightly larger cylinder with a hollow opening. The hollow cylinder will include a smaller cylinder inside it, so that the top section is essentially just two ring cylinders. The inner cylinder will be used to attach a tissue onto. Then the opening of the ostomy bag will be placed into the device, in which a patient can now turn the device in order to effectively clean the opening of the bag. Once done cleaning, the bottom section will include a large button, which a patient can press to eject the dirtied tissue. The client regarded the initial idea as potentially viable as it could be effective in creating distance between the patient and the solid waste in the bag. It would help many patients have an easier time cleaning the opening of the bag. However, it is important to consider the methods in which the tissue will be attached and where the ejected tissue will be going.

Stoma Wrapper

The stoma wrapper would serve as a replacement for the ostomy O-ring that wraps around the stoma. Instead of patients having to cut a circle in order to properly fit the stoma in the bag, this product would have a knob that when turned would wrap soft silicon segments around the stoma that would be fit to the shape of the stoma, similar to a camera aperture. The client was concerned about the device being about to consider people of different sized and shaped stomas.

Because it will only serve ideal-sized stomas, the client was worried about whether it was visible generally, as many patients have different sized stomas.

Discussion

After talking with the client about the various ideas, we formed thoughts about which ideas were viable and could be taken further. Although the lever idea is simple, it creates too much obstruction with the patient and does not account for the lever's contact with the solid waste. The separate cleaning tool had potential as the client said it could positively affect a lot of clients even beyond the people with fine motor disabilities. And finally, the stoma wrapper was concerning for the people with irregular sized stomas or potential damage on the stoma. As a result, we concluded that with some improvements and more specific storyboarding regarding the context and steps behind who this device is used for, the cleaner device could be a viable option for this project.

Appendix F: Instructions for Use

Instructions for Using the Quick-Click Cleaner

The following are the steps to follow when using the Quick-Click Cleaner after waste has been removed from the ostomy bag,

Initial Position

1. The Quick-Click Cleaner's resting position will be upright with the handle on top in the container with the disposable cotton inserts (Figure 23).
2. To start, grab the handle and push the cleaner down into the container to lock in a disposable cotton insert into the cleaner (Figure 24).

Cleaning

1. Lift or remove clothes as necessary to allow easy open access to the opening of the ostomy bag (Figure 25).
2. Grab the handle of the Quick-Click Cleaner and lift out of the container (Figure 26).
3. Open the opening of the ostomy bag by pinching the sides of it with hand (Figure 27).
4. Insert the device into the ostomy bag so the ostomy bag opening is completely wrapped with the disposable cotton insert (Figure 28).
5. Swivel the device around to clean the waste (Figure 29).

Reuse

1. Remove the ostomy bag from the Quick-Click Cleaner (Figure 30).
2. Point the Quick-Click Cleaner towards a trash bin or disposable area (Figure 31).
3. Press the button on the handle to eject the dirty cotton insert (Figure 32).
4. Place the Quick-Click Cleaner back into the container for future use (Figure 33).



Figure 23: Quick-Click Cleaner's resting position



Figure 24: Grabbing the handle



Figure 25: Lifting clothes to access ostomy bag



Figure 26: Lifting the cleaner out of the base



Figure 27: Open the ostomy pouch



Figure 28: Insert the cleaner into bag opening



Figure 29: Swab cleaner in bag



Figure 30: Remove cleaner from bag



Figure 31: Point cleaner towards trash bin



Figure 32: Press button to eject tissue



Figure 33: Reload cleaner for next use

Appendix G: Design Review Summary

During the design review in class on November 2nd, pictures and sketches of the cleaning device were presented. We began by presenting which step of the ostomy pouch emptying process we were seeking to improve—the cleaning of the opening of the pouch—as well as the target user and environment in which the device would be used (bed-ridden people with fine motor disabilities). We then presented the initial sketches of the device as well as two ideas for the wipe application/release mechanism (ejection button and external wipe/tissue cuff) and asked the class, professors, and our shop specialist Bob Taglia to provide feedback on the device in general as well as specific feedback/ideas for the wipe application/release mechanism. We mostly received feedback on the usability/ease of use of the product and suggestions on similar quick release mechanisms.

Table 3: Summary of Feedback from Design Review

Reviewers like	Reviewers dislike	Features to be added	Features to be removed/modified	Additional comments
<u>ACCESS</u> If located on table, easy for bed-ridden person to access Allows person to better access inside of ostomy pouch Shape allows for easier access to crevices of ostomy pouch opening <u>CLEANLINESS</u> User doesn't need to use hands/fingers to touch inside of ostomy pouch	<u>EASE OF USE</u> Many moving parts May be difficult to manipulate device around opening of ostomy pouch <u>USABILITY</u> May not be fully unimanual yet Wipe application and release not fully functional <u>CLEANLINESS</u> How can we ensure that the device won't get soiled upon	Case (so that the device is portable) Stand (so that the device can stand upright on a table)	Wipe application system	<u>USABILITY</u> Where are the tissues/wipes going after they are released? Can tissues be disposed of within the device? Device was designed for wipes; ensure it works with toilet paper/tissue as well May overcomplicate task <u>CLEANLINESS</u> How often would the device need to be cleaned? <u>MANUFACTURING</u> How could this

<u>OTHER</u> Potentially could be used by people without fine motor disabilities	every use?			device be mass produced?
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Table 4: Implementation of Design Review Advice

Suggestion/criticism	Implementation
Many moving parts	Combine stand and wipe applicator such that it is one object
Not fully unimanual*	Continue redesigning shape of object, specifically handle
May be difficult to manipulate device around opening of ostomy pouch*	Continue redesigning shape of object, specifically handle
Wipe application and release not fully functional*	Continue researching quick release options (ex. Fastening system in shop, utility knife, detachable thermometer)
How can we ensure that the device won't get soiled upon every use?	Adjust wipe fastening system such that it is as far away from the top as possible without the wipe falling off

*Solution still needed—immediate action item

Appendix H: Performance Testing

After constructing the prototype, we tested for effectiveness and any possible issues during the use of our device. From the weeks of Nov. 19 - Nov. 29, we tested the device on both ourselves and people who were less familiar with ostomy bags, as we did not have access to people who actually utilize ostomy bags. To recreate both our target audience and realistic scenario, the testers used our device in a bed-like position while only having access to one hand.

Table 5: Prototype Testing and Feedback

Testers	Results
1	“Easy to understand how to use device” “Had difficulty removing cotton insert” “Ejection system doesn’t work”
2	“Device was easy to hold and use” “Needs too much effort to put device into ostomy bag”
3	“Hard to slide cotton insert into bag” “Easy to grip and press button”
4	“Easy to use with one hand” “Simple to use” “Had problem with loading cotton insert into cleaner”
5	“Ejecting needs to work” “Straightforward process”

Feedback

In general, most of our testers easily understood how to use the device based on the design of our prototype. They recognized what each component of our design was meant to function as. The loading/unloading mechanism of the device was simple to use and locating where the bag is inserted into the device had no issues. However, the effectiveness of our prototype was highlighted. Specifically, there were issues regarding consistency in inserting the cotton insert into the bag smoothly and in the ejection system. The cotton insert had some resistance when it was tried to be inserted into the bag utilizing one hand.

Appendix I: Instructions for Construction

Instructions for constructing the Quick-Click Cleaner

Note: See Bill of Materials (Appendix I) for prices and parts.

The Following Materials/Programs are required to build the Quick-Click Cleaner:

- PLA 3d filament
- A minimum 6x6x6 inch 3D printer
- Super Glue
- Cura (or any other software that turns 3d models into g-code)
- Two 1 inch springs

Instructions:

- 1) Begin by downloading the attached stl files [here](#)
- 2) After that is done, you will need to insert each file into cura and slice it to make it ready to print



Figure 34: Cura before slicing a file



Figure 35: Cura after slicing a file

- 3) Print two of every file except the button and storage unit



Figure 36: Image of the empty updated mockup:

- 4) Glue one Spring to the top of the upper divide of the updated model 2 and the bottom of the updated top, and one to the bottom of the lower divide and there, these will act as the release mechanism for the device.



Figure 37: Springs glued on the cross section of the updated mechanism 2

- 5) Glue the updated top together into one piece.



Figure 38: Before gluing the top component together

- 6) Put the components together as illustrated in the image below.

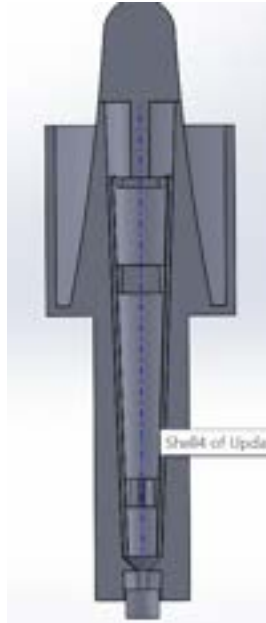


Figure 39: Solidworks model of the completed mockup



Figure 40: Real life counterpart to the complete solidworks assembly

- 7) Glue the two updated mockup 2 pieces together with all the components together as illustrated in the images above and that's everything.



Figure 41: Completed Quick-Click Cleaner

Appendix J: Bill of Materials

Table 5: Bill of Materials

	Item	Distributor	Price
1	PLA 3D Printing Material	Northwestern	\$0
2	Wokape 240Pcs 15 Sizes Compression Springs Assortment Kit, Mini Stainless Steel Extension Springs for Shop and Home Repairs, 0.39" to 2" Length, 0.31" to 0.37" OD, 10mm - 51mm Length, 8-9.5mm OD	Amazon	\$26.78